Delta Hedging with Stochastic Volatility: A Short Model and Data Study

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Abstract

We compare several local hedges for a long call: the Black–Scholes delta, the Heston partial derivative with respect to spot, a minimum-variance Heston hedge, a fitted Hull–White correction, and two-instrument delta–vega positions. A controlled example shows how correlation between spot and variance changes the share hedge. We then examine three close-to-close holding intervals for one SPY call and one AAPL call, and report the saved output of a separate five-underlying notebook experiment. The latter output does not retain its exact observation dates, and its method-specific samples do not support an overall ranking of hedges.

1 Position and local hedge ratios

Take a long call C, short h shares of its underlying S, and, when available, short n units of a second call H. With positions fixed during one holding interval, the change before financing is

$$e = \Delta C - h\Delta S - n\Delta H. \tag{1}$$

The sign convention is useful: a positive delta calls for a positive number of shares to sell short. The European Black–Scholes call with dividend yield q, rate r, maturity τ, and volatility σ has spot delta and vega

$$\Delta_{\mathrm{BS}} = e^{-q\tau}\Phi(d_1), \qquad \mathcal{V}_{\mathrm{BS}} = Se^{-q\tau}\phi(d_1)\sqrt{\tau}, \qquad d_1 = \frac{\log(S/K) + (r - q + \sigma^2/2)\tau}{\sigma\sqrt{\tau}}. \tag{2}$$

Here Φ and ϕ denote the standard normal distribution and density. If σ is inferred from an observed call close, the price is matched by construction. The next close, rather than that same price, tests the hedge.

In Heston's model [2], under the pricing measure,

$$\begin{aligned} dS_t &= (r - q)S_t\,dt + S_t\sqrt{v_t}\,dW_t^S, \\ dv_t &= \kappa(\theta - v_t)\,dt + \xi\sqrt{v_t}\,dW_t^v, \qquad d\langle W^S, W^v\rangle_t = \rho\,dt. \end{aligned} \tag{3}$$

The partial derivative C_S holds instantaneous variance v fixed. It need not be the best one-stock hedge when variance moves with the stock. To first order, $dC \simeq C_S dS + C_v dv$; minimizing $\mathrm{Var}(dC - h\,dS)$ therefore gives

$$\Delta_{\mathrm{MV}} = \frac{\mathrm{Cov}(dC, dS)}{\mathrm{Var}(dS)} = C_S + \frac{\rho\xi}{S}C_v. \tag{4}$$

C_v differentiates with respect to *variance*, whereas $\mathcal{V}_{\mathrm{BS}}$ in (2) differentiates with respect to implied *volatility*. Equation (4) removes the component of the variance shock that is locally correlated with the stock shock; it does not remove all variance risk.

The empirical Hull–White correction [3] takes the form

$$\Delta_{\mathrm{HW}} = \Delta_{\mathrm{BS}} + \frac{\mathcal{V}_{\mathrm{BS}}}{S\sqrt{\tau}}(a + bd + cd^2), \qquad d = \Delta_{\mathrm{BS}} \quad \text{for a call,} \tag{5}$$

where a, b, c are fitted on earlier hedging errors. This is an estimated adjustment, not the Heston derivative C_S. With a longer-dated call H, a Heston delta–vega position can instead cancel both the variance and stock sensitivities by choosing

$$n = \frac{C_v}{H_v}, \qquad h = C_S - nH_S. \tag{6}$$

The analogous Black–Scholes construction replaces variance sensitivities by Black–Scholes vegas and uses Black–Scholes deltas for both calls.

2 A prescribed one-day shock

The notebook in [1] sets $S_0 = K = 100$, $\tau = 0.25$, $r = 0.03$, $q = 0.013$ and $(v_0, \kappa, \theta, \xi, \rho) = (0.04, 2, 0.04, 0.6, -0.7)$. Its Heston call price is 3.9234; the price-matched Black–Scholes volatility is 18.70%. The resulting deltas are $\Delta_{\mathrm{BS}} = 0.535$, $C_S = 0.628$, and $\Delta_{\mathrm{MV}} = 0.461$. In particular, the negative ρ and positive C_v explain why the last ratio is below C_S. Fitting (5) on the first 100 paths of a 200-path simulated training sample gives $\Delta_{\mathrm{HW}} = 0.464$.

The notebook reprices the calls after one trading day under $\Delta S = \pm 1$ and $\Delta v = (\rho\xi/S_0)\Delta S = \mp 0.0042$. Table 1 reports (1) per call share; its one-year call quantities are n. These are computed scenario errors, not realized trading returns.

Table 1: Hedge positions and errors under the two prescribed shocks.

Hedge	Shares h	Calls n	e ($S\uparrow$)	e ($S\downarrow$)
Unhedged	0	—	+0.451	−0.471
Black–Scholes delta	0.535	—	−0.084	+0.064
Heston C_S	0.628	—	−0.177	+0.158
Heston Δ_{MV}	0.461	—	−0.010	−0.010
Hull–White Δ_{HW}	0.464	—	−0.013	−0.006
BS delta–vega	0.246	0.511	−0.049	+0.036
Heston delta–vega	−0.050	1.014	−0.002	−0.003

The minimum-variance delta and the second Heston call substantially reduce the errors for these *particular* shocks. The simulated parameters violate the Feller inequality: $2\kappa\theta = 0.16 < \xi^2 = 0.36$. Neither the fitted correction nor the shock table is an independent out-of-sample market test.

3 Four observed SPY and AAPL closes

The slides and notebook record Yahoo Finance closing observations on October 30 and November 2, 3, and 4, 2026. They follow a SPY call with $K = 760$ and November 5 expiry and an AAPL call with $K = 345$ and November 6 expiry. There are exactly three entry-to-next-close intervals per call: October 30–November 2, November 2–3, and November 3–4. The November 4 delta is observed but has no subsequent close in the sample. These figures are transcribed observations, not newly downloaded quotations.

Each entry close determines a European Black–Scholes implied volatility and delta with $r = q = 0$; $h_t = \text{round}(100\Delta_{\text{BS}})$ shares are shorted against one long 100-share call. On each interval the gross dollar error is

$$E_t^{\text{gross}} = 100(C_{t+1} - C_t) - h_t(S_{t+1} - S_t). \tag{7}$$

For reference, a hypothetical annual stock-loan rate $b = 1\%$ would produce $E_t^{\text{net}} = E_t^{\text{gross}} - h_t D_t - bh_t S_t d_t/365$, where D_t is the cash dividend per share and d_t is the number of *calendar* days between closes. No cash dividend occurs in these stored intervals; the reported comparison uses gross errors and excludes the assumed loan fee.

Table 2: Recorded entry deltas and next-close gross errors, dollars per call.

Call	Holding interval (2026)	Entry Δ_{BS}	Short shares h_t	E_t^{gross}
SPY	Oct. 30–Nov. 2	0.4036	40	−43.20
SPY	Nov. 2–3	0.3043	30	−67.50
SPY	Nov. 3–4	0.3366	34	+125.36
AAPL	Oct. 30–Nov. 2	0.5052	51	+21.81
AAPL	Nov. 2–3	0.3834	38	−244.04
AAPL	Nov. 3–4	0.0869	9	+779.51

For each call, define $\text{RMSE}(E) = \sqrt{\sum_{t=1}^{3} E_t^2/3}$; the unhedged series is $100\Delta C_t$. The stored comparison gives SPY unhedged and hedged RMSEs of \$335.03 and \$85.90, against an initial premium of \$477. AAPL gives \$634.56 and \$471.76, against \$825. Thus $1 - \sum(E_t^{\text{gross}})^2 / \sum(100\Delta C_t)^2$ is 93.4% for SPY and 44.7% for AAPL; hedged RMSE divided by the initial premium is 18.0% and 57.2%, respectively. In AAPL's last interval the stock moves from 331.90 to 349.51 across the strike, while the call moves from 0.37 to 9.75. The entry hedge of nine shares leaves the \$779.51 residual in Table 2. Three intervals cannot support a general preference for hedging one underlying over the other.

4 Saved five-underlying comparison

A separate notebook section selects calls on SPY, QQQ, AAPL, MSFT, and TSLA from a then-current option chain, queries approximately 90 calendar days of price history, and estimates Heston parameters from daily option surfaces. It fits (5) on earlier eligible dates and scores later dates. Table 3 describes the saved sample: a fitted date has an accepted Heston calibration; a paired date has both entry and next closes; an option-day is one usable option on such a pair. Median quote counts are over eligible quote dates, not just dates with accepted Heston fits. The later-date column is assigned before missing values are removed separately for the scored methods.

Table 3: Saved five-underlying sample audit. Second-call availability is a fraction of option-days.

Underlying	Fitted dates	Median quotes/date	Paired dates	Option-days	Later dates	Second call (%)
SPY	18	12	17	322	9	0
QQQ	9	8	8	147	4	0
AAPL	54	19	53	971	27	66
MSFT	17	10	16	296	8	74
TSLA	22	13	21	340	11	88

For method m the notebook's financed one-period error per option share is

$$e_{m,t} = \Delta C_t - h_{m,t}\Delta S_t - n_{m,t}\Delta H_t - r_t\Delta t_t(C_t - h_{m,t}S_t - n_{m,t}H_t) - h_{m,t}D_t. \tag{8}$$

Its reported gain $G_m = 1 - \sum_{t\in I_m} e_{m,t}^2 / \sum_{t\in I_m} e_{\mathrm{BS},t}^2$ uses the same rows I_m for method m and its Black–Scholes benchmark. Crucially, I_m can differ *across* methods because some prices or a second call are unavailable. Table 4 reproduces saved percentages, not newly computed results.

Table 4: Saved gain in squared-error loss versus Black–Scholes delta, in percent; each method has its own matched rows.

	Heston C_S	Heston Δ_{MV}	Hull–White	Heston delta–vega
SPY	−98.2	+58.4	−12.4	—
QQQ	−65.2	+22.4	−47.3	—
AAPL	+13.0	−20.4	+6.6	−374.1
MSFT	−0.1	−28.9	−6.9	+52.5
TSLA	−24.3	−111.1	+8.0	−39.5

For example, SPY's +58.4% says that its Heston minimum-variance hedge had 41.6% of the Black–Scholes squared error on the rows used in that comparison. A negative gain means greater squared error; it is not a negative return on SPY. There is no usable second hedge call for SPY or QQQ in this saved panel. In particular, the +52.5% MSFT delta–vega figure cannot be compared with another method's gain as if their test rows were identical.

The five-underlying archive does *not* preserve the run timestamp, exact quote dates, or selected option symbols. Its observations therefore must not be identified with the four dated closes in Section 3. Moreover, the listed calls are American-style contracts while the pricing and hedge formulas here are European approximations. The brief window, missing hedge calls, changes in option liquidity, and uneven Heston calibration all limit interpretation. A prospective comparison would first fix common contracts, dates, and scored rows for every eligible hedge, then account for option spreads, stock borrow costs, and early exercise.

References

[1] A. Adimurthi, *Delta Hedging*, slide deck and accompanying five-ticker notebook, Erdős Institute, Quantitative Methods in Finance, Fall 2026 (available on github).

[2] S. L. Heston, "A closed-form solution for options with stochastic volatility with applications to bond and currency options," *Review of Financial Studies* **6** (1993), 327–343.

[3] J. Hull and A. White, "Optimal delta hedging for options," *Journal of Banking & Finance* **82** (2017), 180–190.